CascadeOS predicts infrastructure cascades before cities lose control.

One operating view for NYC water, transit, health, emergency response, live camera signals, rerouting, and city briefings.

40	4	26	6
NYC infrastructure nodes	domain agents	nodes in demo cascade	completed build phases

Built by Raunak Choudhary as solo team Core Catalyst.

Say this: CascadeOS is a command center for the moments when one city failure becomes many.

Infrastructure failures do not stay inside one agency.

A water incident can become a transit delay, a hospital access issue, and an emergency routing problem within minutes.

TODAY

Siloed screens

- Water, MTA, hospitals, and responders see separate dashboards.
- Teams react after impact becomes visible.

RISK

Hidden dependencies

- The critical part is not one failure.
- The critical part is what the failure touches next.

NEED

Shared prediction

- Officials need a single answer fast.
- What failed, what is next, and what should move now.

Say this: The project focuses on the gap between agencies, because cascades happen in that gap.

CascadeOS turns live city signals into operational decisions.

HOW IT WORKS

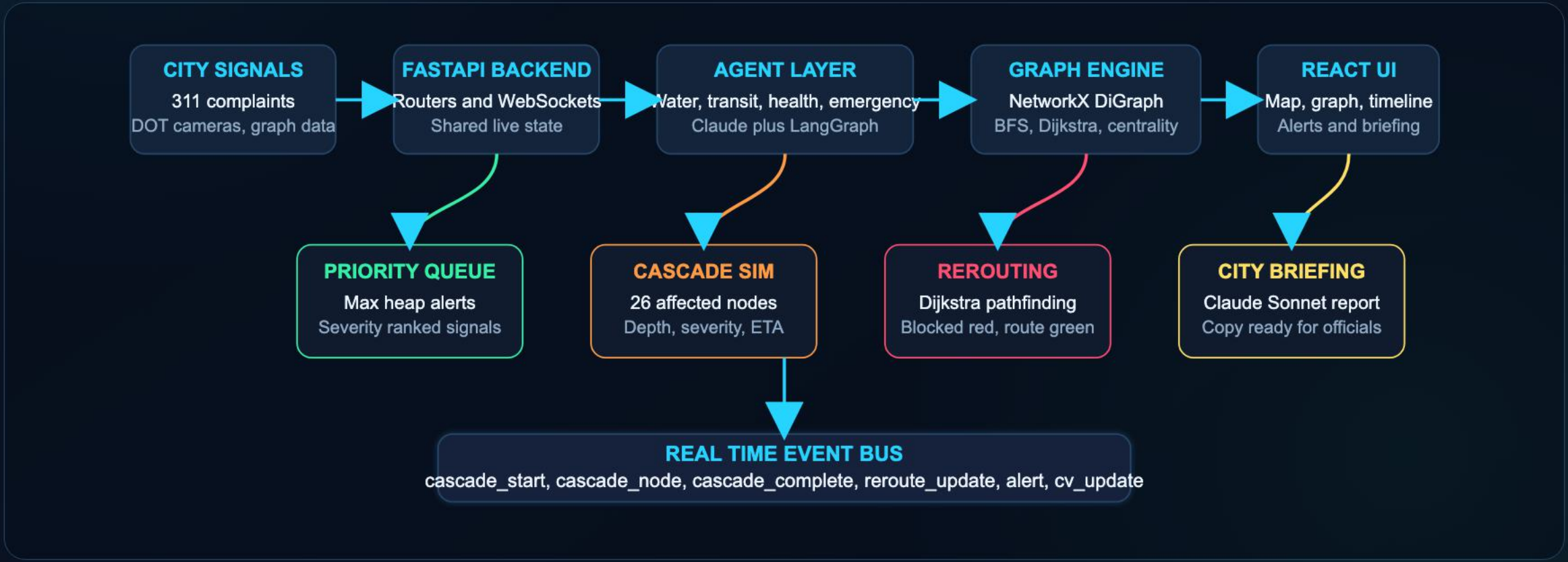
- Model NYC systems as a NetworkX directed graph.
- Run domain agents for water, transit, health, and emergency response.
- Predict cascade spread with weighted BFS.
- Route responders around blocked nodes with Dijkstra.

WHAT USERS GET

- Live NYC map with affected infrastructure.
- Agent reasoning and priority alerts.
- Cascade timeline and what-if simulation.
- Claude generated city briefing for officials.

Say this: The system does not just show data. It converts data into a predicted impact chain and a recommended response.

Architecture built for real time cascade intelligence.



Say this: Signals enter through FastAPI, agents reason over them, graph algorithms predict impact, and WebSockets push decisions to the UI.

The judge demo is one incident, one minute, many systems.

01
Trigger 34th Street water main failure.

02
Weighted BFS predicts cascade across NYC graph.

03
26 infrastructure nodes become affected.

04
Agents publish alerts and reasoning.

05
Dijkstra finds emergency reroute.

06
Claude generates city briefing.

Say this: The demo shows prediction, prioritization, response, and communication in one flow.

What is working right now.

MAP

- NYC infrastructure nodes with detail panel.
- Cascade and reroute layers.

AGENTS

- Four live domain agents.
- Shared alert priority queue.

SIMULATION

- Weighted BFS cascade engine.
- Timeline and what-if controls.

ROUTING

- Dijkstra emergency reroute.
- Blocked path red, reroute green.

BRIEFING

- Claude Sonnet city report.
- Full structured copy action.

VISION

- NYC DOT camera feeds.
- YOLOv8 signal pipeline with fallback.

Say this: This is not a mockup. These are clickable, running features in the current app.

The core is DSA plus agents, not just prompts.

ALGORITHMS

- NetworkX directed graph with 40 infrastructure nodes.
- Betweenness centrality for node importance.
- Weighted BFS for cascade spread.
- Dijkstra for blocked-node rerouting.
- Max heap for alert priority.

AI LAYER

- Claude Sonnet for agent reasoning and city briefings.
- LangGraph orchestration across domain agents.
- YOLOv8 for traffic camera anomaly signals.
- Fallback detection keeps the demo reliable.

Say this: The AI is useful because it sits on top of real graph algorithms and live state.

CascadeOS helps cities move from reaction to readiness.

FASTER AWARENESS

- Officials see downstream impact before every agency calls it separately.

BETTER COORDINATION

- Water, transit, health, and emergency response share one incident picture.

CLEARER COMMUNICATION

- The city briefing turns technical state into a public safety summary.

Say this: The value is not replacing humans. It is giving responders a shared prediction while there is still time to act.

Full stack system built for a live hackathon demo.

BACKEND

- FastAPI
- Python 3.12
- NetworkX
- WebSockets

INTELLIGENCE

- Claude Sonnet
- LangGraph
- Weighted BFS
- Dijkstra
- YOLOv8

FRONTEND

- React and Vite
- deck.gl and Mapbox
- React Flow
- Dark and light themes

Say this: The project is end to end: data, intelligence, algorithms, real time transport, and user interface.

Future work turns the proof of concept into a city resilience platform.

NEAR TERM

- Deployment hardening and persistent incident storage.
- More real data feeds across utilities and emergency services.
- Scenario library for storms, heat, water, transit, and grid events.

LONG TERM

- Temporal GNN forecast for 24 hour cascade risk.
- 3D digital twin view of the city.
- Natural language simulation for officials.
- Multi-city version for resilience planning.

Say this: The current version is reactive prediction. The next version forecasts before the incident happens.